

Question:

The y -intercepts, P and Q , of two perpendicular lines intersecting at the point $A(6,8)$ have a sum of zero. What is the area of $\triangle APQ$?

- (A) 45 (B) 48 (C) 54 (D) 60 (E) 72

Reasoning + Answer:

To solve the problem, we need to understand the properties of the lines and their y -intercepts. The lines are perpendicular and intersect at the point $A(6,8)$. Let's denote the slopes of the lines as m_1 and m_2 . Since the lines are perpendicular, $m_1 \cdot m_2 = -1$.

The equations of the lines can be written as:

$$y = m_1(x - 6) + 8 = m_1x - 6m_1 + 8$$

$$y = m_2(x - 6) + 8 = m_2x - 6m_2 + 8$$

The y -intercepts of the lines are the values of y when $x = 0$. So, the y -intercepts are:

$$P = -6m_1 + 8$$

$$Q = -6m_2 + 8$$

Since $m_1 \cdot m_2 = -1$, we can use the fact that $m_2 = -\frac{1}{m_1}$. Therefore, the sum of the y -intercepts is:

$$P + Q = -6m_1 + 8 + (-6m_2 + 8) = -6m_1 - 6\left(-\frac{1}{m_1}\right) + 16 = -6m_1 + \frac{6}{m_1} + 16$$

Given that $P + Q = 0$, we have:

$$-6m_1 + \frac{6}{m_1} + 16 = 0$$

Multiplying through by m_1 :

$$-6m_1^2 + 6 + 16m_1 = 0$$

Rearranging terms:

$$-6m_1^2 + 16m_1 + 6 = 0$$

Let $u = m_1$, then the equation becomes:

$$-6u^2 + 16u + 6 = 0$$

Dividing by -2 :

$$3u^2 - 8u - 3 = 0$$

Using the quadratic formula $u = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ where $a = 3$, $b = -8$, and $c = -3$:

$$u = \frac{8 \pm \sqrt{64 + 36}}{6} = \frac{8 \pm 10}{6}$$

So, the solutions are:

$$u = \frac{18}{6} = 3 \quad \text{or} \quad u = \frac{-2}{6} = -\frac{1}{3}$$

Therefore, $m_1 = 3$ or $m_1 = -\frac{1}{3}$. If $m_1 = 3$, then $m_2 = -\frac{1}{3}$.

If $m_1 = -\frac{1}{3}$, then $m_2 = -3$.

Now, we can find the y -intercepts:

$$P = -6(3) + 8 = -18 + 8 = -10$$

$$Q = -6\left(-\frac{1}{3}\right) + 8 = 2 + 8 = 10$$

Or:

$$P = -6\left(-\frac{1}{3}\right) + 8 = 2 + 8 = 10$$

$$Q = -6(3) + 8 = -18 + 8 = -10$$

So,